PROJECT REPORT ON CARRYING OUT SURVEY TECHNIQUES FOR A UNIVERSITY CAR PARKING AREA

Submitted in the partial fulfillment of the Requirements for the award of the degree of Bachelor of Technology In Civil Engineering

By

EssayCorp 5 years

Department of Civil engineering

ABSTRACT

Importance of Surveying to Civil Engineers

The planning and design of all Civil Engineering projects such as construction of highways, bridges, tunnels, dams etc are based upon surveying measurements. Moreover, during execution, project of any magnitude is constructed along the lines and points established by surveying.

Thus, surveying is a basic requirement for all Civil Engineering projects.

Other principal works in which surveying is primarily utilized are

- to fix the national and state boundaries;
- to chart coastlines, navigable streams and lakes;
- to establish control points;
- to execute hydrographic and oceanographic charting and mapping
- to prepare topographic map of land surface of the earth.

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OBJECTIVES OF SURVEYING

- To collect field data;
- To prepare plan or map of the area surveyed.
- To analyze and to calculate the field parameters for setting out operation of actual engineering works.
- To set out field parameters at the site for further engineering works.

CONTENTS

1. AIM OF THE PROJECT

2. INTRODUCTION
3. RECONAISSENCE SURVEY
4. TOPOGRAPHICAL SURVEY
5. FIXING OF SITE BOUNDRIES BY LINEAR SURVEY
6. CONTROL POINTS BY LEVELLING INSTRUMENT
7. VERTICAL LEVELLING BY HOI & RA ISE & FALL
8) DRAIN PROFILING
9. TWO PEG TEST
10. CONCLUSION

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AIM OF THE PROJECT

- To Understand the principles of site surveying
- Be able to use site surveying instruments
- Understanding cartographic detailing of construction works
- Understand the Software for use in surveying

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LOCATION MAP

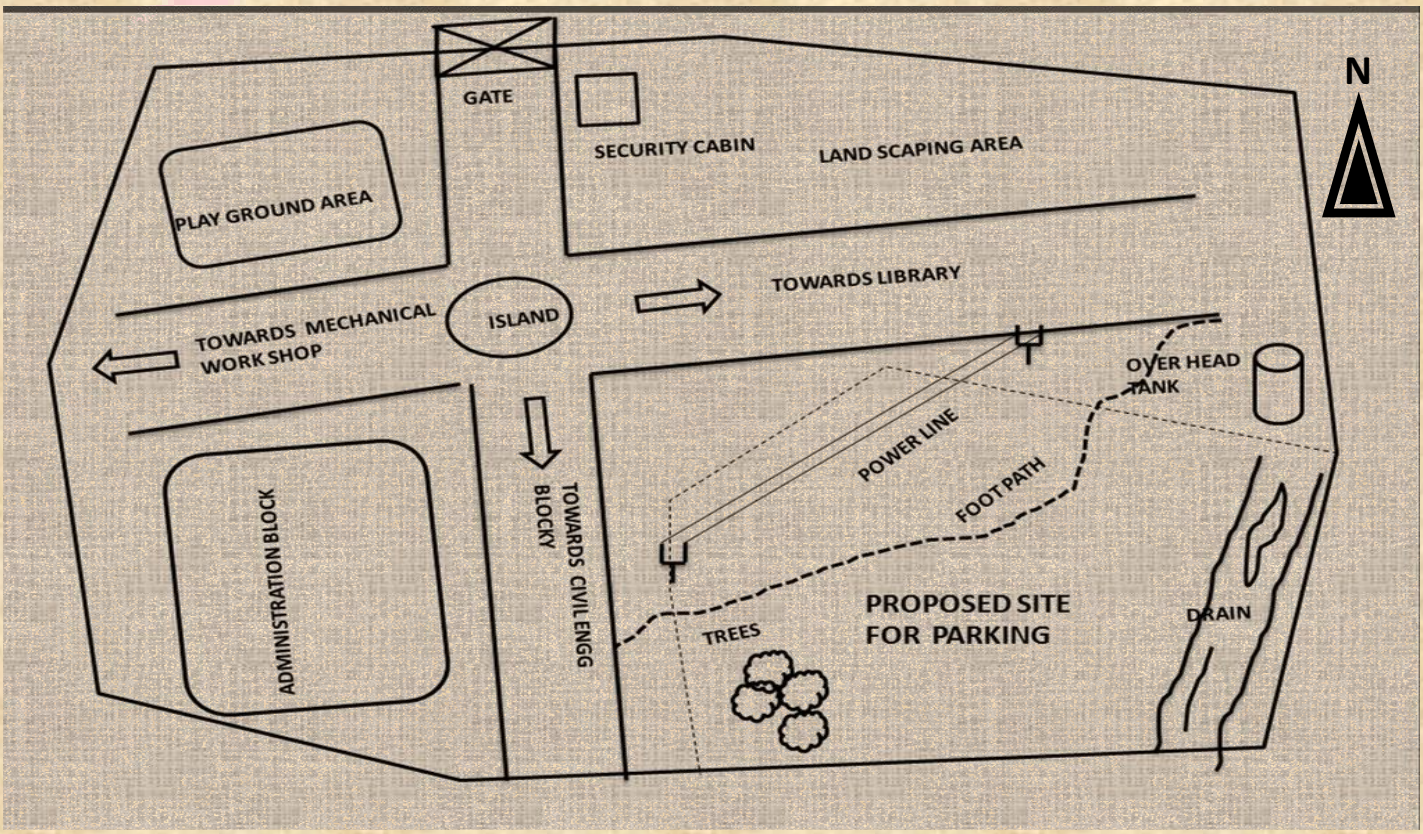
RECONNAISANCE SURVEY

The site visit is the first gear in the design process. A detailed site analysis has to be done so as to understand the features of the site, which will be very important during the design. The purpose of the site analysis is to record and evaluate information on the site and its surroundings, and to use this evaluation in the design response.

Following are the points to be considered while selecting the site.

- Location
- Orientation
- Wind Direction
- Soil Type & Condition
- Topography
- Vegetation & Natural Features
- Surrounding Land Uses & Buildings
- Locally Available Resources

A SKETCH SHOWING THE TOPO DETAILS OF THE PROPOSED SITE



FIXING OF BOUNDARIES BY LINEAR SURVEY METHOD

LINEAR SURVEYING:

The determination of the distance between two points on the surface of the earth is one of the basic operations of surveying. Measurement of horizontal distances or measuring linear measurement is required in chain surveying, traverse surveying and other types of surveying

- Methods of making linear measurements
- Direct methods
- Optical methods
- E.D.M methods

For the proposed site we are following direct method:

Direct Method:

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In direct method, the distance is actually measured during field work using a chain or a tape. This is the most commonly used method for linear measurements.

FIELD MEASUREMENT DATA:

DATE:
TIME:
WEATHER:
LOCATION:
TYPE OF SURVEY: linear surveying by direct method
CREW MEMBERS:
EQUIPMENTS: Metric Chain-30m,Tape,Arrows,Ranging Rods, Offset Rods,Pegs,Plumb bob, Line Ranger Cross Staff.

PROCEDURE:

- Establishing Control Points spreading entire area of site.
- As we have clear vicinity of entire site a base line of max diagonal length of the site to be marked on the ground by doing ranging procedure.
- Keep the ranging rods in line across the diagonal length.
- Mark the line with lime powder on ground.
- Mark all the boundaries of the site by placing Ranging Rods.
- Spread the chain along the base line marking.
- Place the cross staff along the chain length, from which the locations of ranging rods of the boundaries shall be visible.
- Measure the distances parallel & perpendicular to the point of cross staff to the boundary points with a measuring tape.
- Repeat the procedure along the base line till we complete all the measurements of boundaries & near by topo details.
- Record the data in a Field Measurement Book with all relevant drawings & area calculations.
- By using triangulation methods mention the angular measurements of the site boundaries also.

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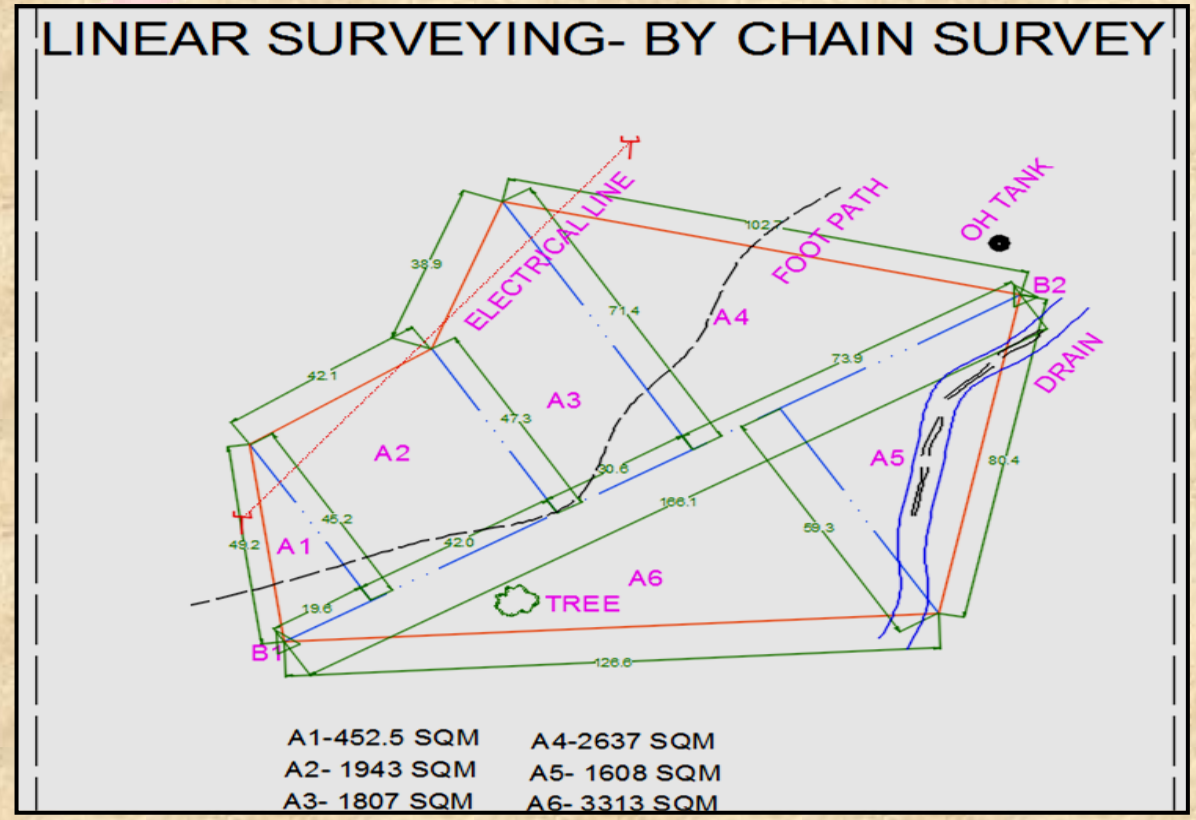
PLOTTING BY USING AUTOCAD : Annexure Attached CAD FILE.



CALCULATIONS:

All area calculations have been done by using Autocad Software.
In general by using the principles of triangulations also we can calculate the areas.

B.VERTICAL CONTROL LEVELLING:

HEIGHT OF COLLIMATION METHOD:

It consist of finding the elevation of the plane of collimation (H.I.) for every set up of the instrument, and then obtaining the reduced level of point with reference to the respective plane of collimation.

Elevation of plane of collimation for the first set of the level determined by adding back side to R.L. of B.M.
2. The R.L. of intermediate point and first change point are then obtained by starching the staff reading taken on respective point (IS & FS) from the elation of the plane collimation. [H.I.]
3. When the instrument is shifted to the second position a new plane collimation is set up. The elevation of this plane is obtained by adding B.S. taken on the C.P. From the second position of the level to the R.L. C.P. The R.L. of successive point and second C.P. are found by subtract these staff reading from the elevation of second plane of collimation Arithmetical check

Sum of B.S. – sum of F.S. = last R.L. – First R.L.

FIELD MEASUREMENT DATA:

DATE:
TIME:
WEATHER:
LOCATION:
TYPE OF SURVEY: Leveling by Height Of Collimation Method.
CREW MEMBERS:
EQUIPMENTS: Auto Level, Staff, Metric Chain-30m, Tape, Arrows, Ranging Rods,

Offset Rods, Pegs, Plumb bob.

PROCEDURE:

- Check the calibration of the Auto Level before leveling starts.
- Fix a Temporary Bench mark (TBM) at a little higher elevation of the site.
- Consider control points as the starting & end point of Base line. B1 & B2.
- Spread the chain & mark the arrows at every 10m interval along the base line.
- Similarly mark the Tie lines from the 10m markings to make a grid of 5x10m.
- Set up the Auto level over Control Point B1.
- Suspend the Plumb Bob from the center of the tripod for positioning the tripod approximately over the inserted peg/arrow on the ground.
- Make the bubble to the center of the glass tube by adjust the foot screws.
- Put the Staff over the TBM & take the Reading.
- Record it in BS.
- Then Height of Collimation= TBM RL+ Staff Reading.
- Enter the reading of HOI.
- Place the staff at 10m distance marking & note the staff reading.
- Enter this value in FS/IS.
- RL of that point will be HOI-FS=RL of that point.
- Repeat the procedure along & across the lines till we have clear vision of the staff reading.
- Shift the instrument to next convenient place. Record as change point CP1
- Before shifting the instrument the last reading to be entered in FS.
- Repeat the procedure till we reach the control point B2.
- Tabulate the readings precisely in a Field Measurement book.
- Check for closing error & should not be more than **5SQRTn.**

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PLOTTING BY USING AUTOCAD :

Enclosures Attached-CAD FILE.

CALCULATIONS :

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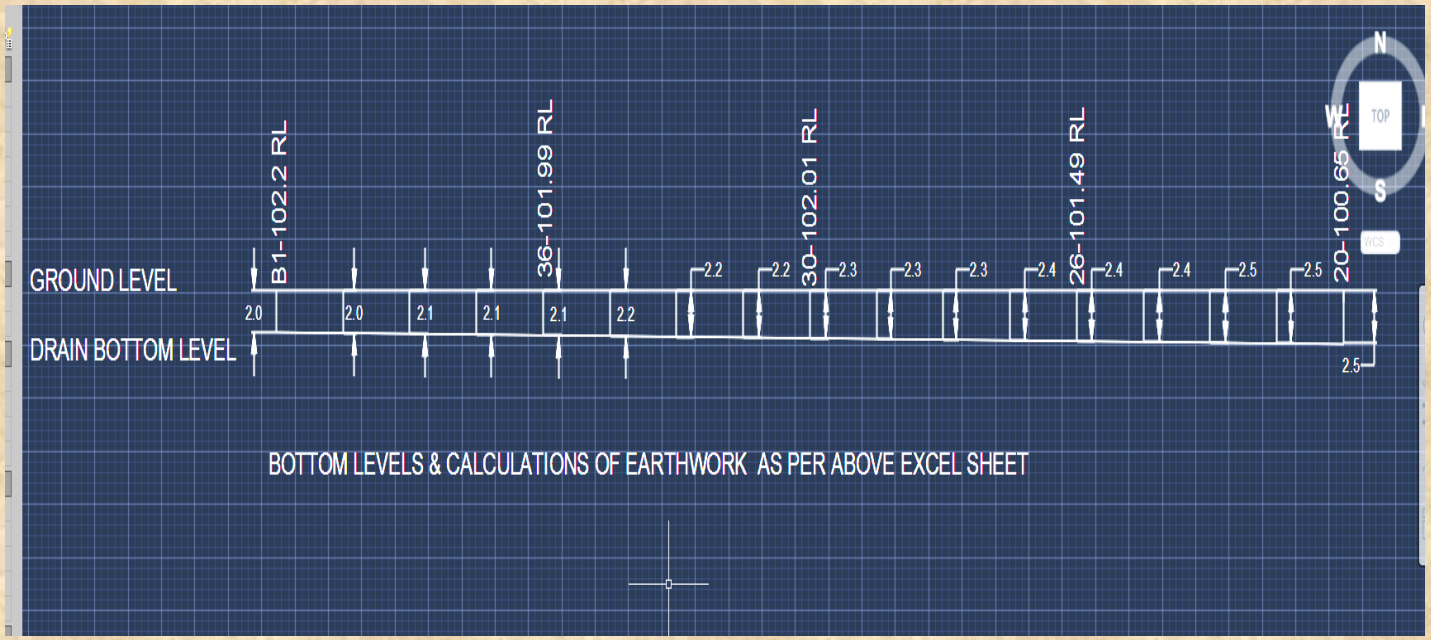
HEIGHT OF COLLIMATION METHOD							RISE	FALL
POINT ID	RL	BS	IS	FS	HI	REMARKS		
TBM	100	2.85			102.85			
B2	99.84		3.01		102.85	CONSIDERING B2 AS STARTING POINT		-0.16
1	100.09		2.76		102.85		0.25	
2	100.06		2.79		102.85			-0.03
3	99.86		2.99		102.85			-0.2
4	99.8			3.05	102.85			-0.06
		2.21			102.01	CP-1		
5	98.78		3.23		102.01			-1.02
6	98.87		3.14		102.01		0.09	
7	99.04		2.97		102.01		0.17	
8	99.46		2.55		102.01		0.42	
9	99.34		2.67		102.01			-0.12
10	99.22			2.79	102.01			-0.12
		3.54			102.76	CP-2		
11	99.41		3.35		102.76		0.19	
12	99.58		3.18		102.76		0.17	
13	99.88		2.88		102.76		0.3	
14	100.14		2.62		102.76		0.26	
15	99.89		2.87		102.76			-0.25
16	99.81		2.95		102.76			-0.08
17	99.53			3.23	102.76			-0.28
		3.68			103.21	CP-3		
18	100.25		2.96		103.21		0.72	
19	100.43		2.78		103.21		0.18	
20	100.65		2.56		103.21		0.22	
21	100.76		2.45		103.21		0.11	
22	100.85		2.36		103.21		0.09	
23	100.89			2.32	103.21		0.04	
		2.77			103.66	CP-4		
24	101.3		2.36		103.66		0.41	
25	101.41		2.25		103.66		0.11	
26	101.49		2.17		103.66		0.08	
27	101.47		2.19		103.66			-0.02
28	101.61			2.05	103.66		0.14	
		2.35			103.96	CP-5		
29	101.74		2.22		103.96		0.13	
30	102.01		1.95		103.96		0.27	
31	102.07		1.89		103.96		0.06	
32	101.91		2.05		103.96			-0.16
33	101.73		2.23		103.96			-0.18
34	102.11		1.85		103.96		0.38	
35	102.19		1.77		103.96		0.08	
36	101.99		1.97		103.96			-0.2
37	101.64		2.32		103.96			-0.35
B1	102.2			1.76	103.96	CONTROL POINT B2	0.56	
	2.2	17.4		15.2			5.43	-3.23
CHECK	?BS-? FS		LRL-FRL	?RAISE-? FALL				
	2.2		2.2	2.2				

DRAIN PROFILING:

DRAIN PROFILING							
	POINT	RL	CUT(+)/Fill(-)	DBL	AREA AT SECTIONS	VOLUME	
0	B1	102.2	2	100.2	16	0	
5		102.15	2	100.15	16	80	
10		102.1	2.1	100	16.8	82	
15		102.05	2.1	99.95	16.8	84	
20		101.99	2.1	99.89	16.8	84	
25		102	2.2	99.8	17.6	86	
30		102.01	2.2	99.81	17.6	88	
35		102	2.2	99.8	17.6	88	
40		102.01	2.2	99.81	17.6	88	
45		101.837	2.3	99.537	18.4	90	
50		101.667	2.3	99.367	18.4	92	
55		101.487	2.4	99.087	19.2	94	
60		101.49	2.4	99.09	19.2	96	
65		101.36	2.4	98.96	19.2	96	
70		101.11	2.5	98.61	20	98	
75		101.02	2.5	98.52	20	100	
80		100.65	2.5	98.15	20	100	
							CUM
					CUTTING	1446	



Attached Enclosures: CAD FILE

PROCEDURE:

- Considered chainage from B1 point to Point No-20. TOTAL-80m
- For a length of 80m with 1:150 slope arrived a depth of 0.53m
- Calculated the vertical distances from the Ground Level based on the slope given

Ex: For 150M Distance-1M Down SLOPE GIVEN

Then for 80M-0.53

- Similarly for every 5m Interval the same calculations has been carried out for getting the
- Drain Bottom Levels from GL.

- After arriving the vertical heights from GL CUT/Fill calculations are carried out.

Example:

	POINT	RL	CUT(+)/Fill(-)	DBL	AREA AT SECTIONS	VOLUME	

5		102.15	2	100.15	16	80	

At Section 5 the depth of cutting is 2m from GL.

Therefore Volume calculation is: **Cross Sectional Area at that section X Length of that section**
The same procedure is continued for all the sections till we reach the end chainage.
As per the profile of the grounf condition we have encountered all Cutting only.
The Volume of Cutting comes-1446 Cum.

Procedure for Setting out the Drain:

Mark the line on the ground by using chain up to 80m.
Put Cross staff at B1 Point .
Given The width of the drain is 8m
Stake out a perpendicular of 4m on either sides by using the cross staff.
Shift the cross staff position the 20m intervals and repeat the sam procedure on both of the center line.
Continue the same process till the end of the length of the drail.
Mark with arrows or Ranging Rods for the perpendicular distances marked by using cross staff.
Using lime powder join the lines on either side of the center line.
Check the width & length of the markings once it is done.

TWO PEG METHOD:

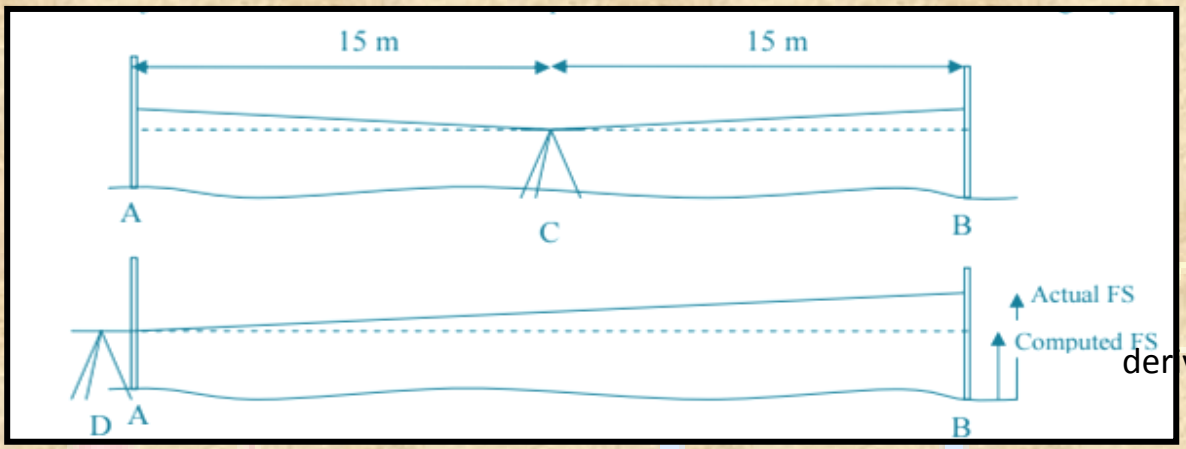
This method is either for an optical or digital level, or a transit being used as a level. If thiserror is corrected with a transit, it also improves the accuracy of its vertical angle readings.
The two-peg test is very simple, but provides a way to test the accuracy of a level, and if you know which screw to turn (for analog instruments) or menu to follow (for the digital level), you can adjust it to remove the error. See specific instrument instructions for making adjustments.

PROCEDURE:

In reasonably level terrain, mark out two rod positions (A and B) on stable surfaces about 30 m apart, and set up the instrument exactly midway between them at C. A good place is along a sidewalk (preferably in a shaded area, or

on an overcast day, to avoid optical problems from
heat-derived air turbulence), using sidewalk chalk to mark

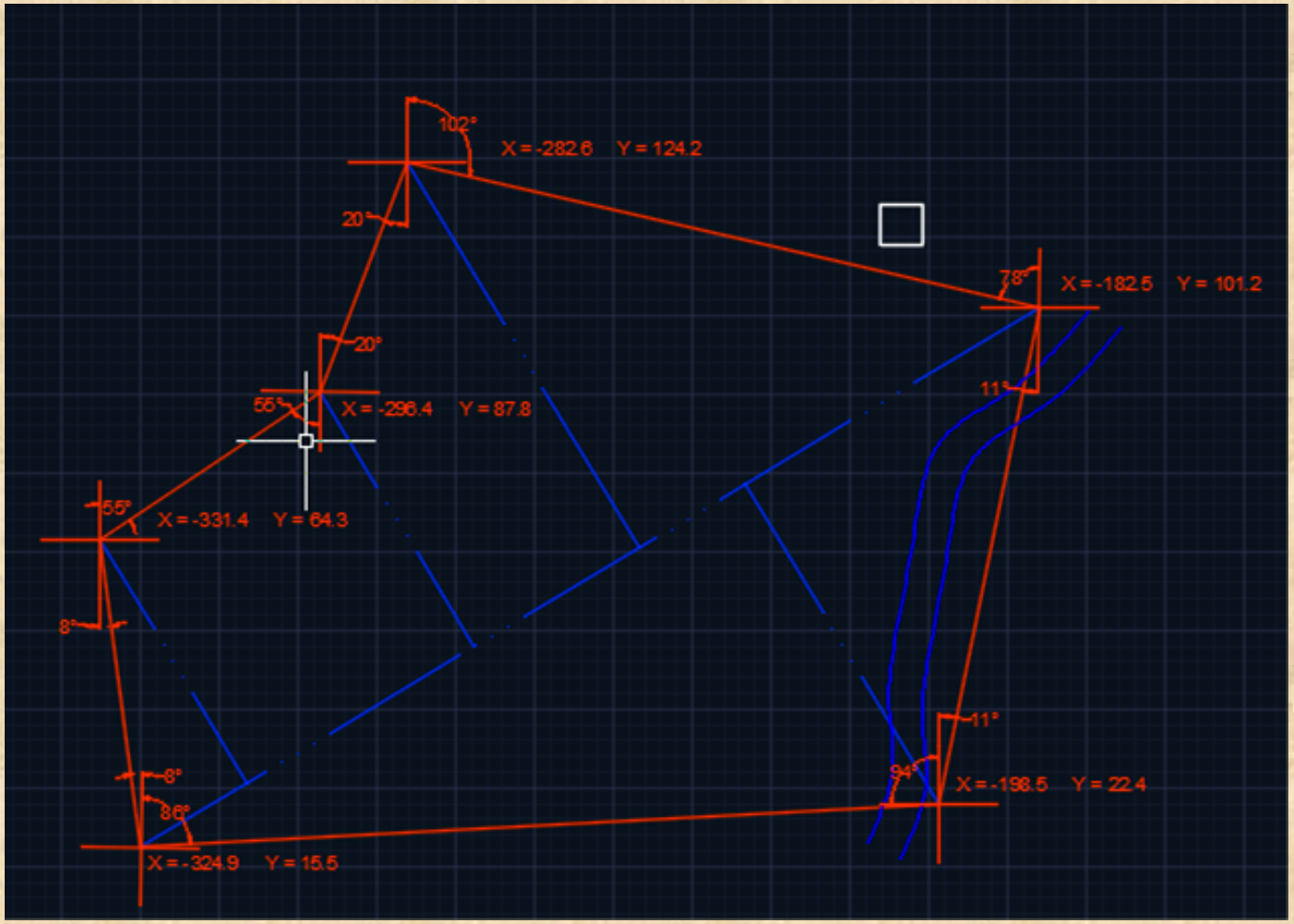
your points. 2. Sight to A to get a rod reading, then to B to get a reading. Assign zero to A, and



derive B as A-B. Say we get 1.524 at A and 1.501 at B, the elevation at B is 0.023. (It doesn't matter what the elevation above sea level is – we only need what its elevation is relative to A.) 3. Move the instrument to ~3m from position A, and get a new reading to A. Subtract the elevation at B (0.023) to get the foresight you should expect to get. Say your reading at A is 1.257:
FS computed = 1.257 – 0.023 = 1.234 4. Now sight to B to get a reading, and compare it to the expected reading. A C B D A B 15 m 15 m Computed FS Actual FS 5. If there's an error, your FSactual will be different from this, and requires an adjustment. Your task is to move it up or down until you get the computed FS.

ANGULAR MEASUREMENTS:

Bearing calculations are done by Autocad Software.



Attached Enclosures.

HORIZONTAL ANGLES

GIVEN CODING FOR EACH CORNER OF THE BOUNDARY AS B1,A1,B2,A2,A3,A4

Inst. Station	Angle	Face	Repetition	Circle Reading	Angle value	Mean
B1	<A1B1C4	L	1	27,55,3	27,55,5	27,55,5
			2	55,50,5		
		R	1	27,55,4	27,55,3	
			2	55,50,5		
A1	B2A1B1	L	1	104,34,45	104,34,47	104,34,46
			2	209,9,30		
		R	1	104,34,44	104,34,45	
			2	209,9,29		
B2	A1B2A2	L	1	91,29,5	91,29,6	91,29,6
			2	182,58,10		
		R	1	91,29,4	91,29,7	
			2	182,58,11		
A2	B2A2A3	L	1	97,50,03	97,50,03	97,50,03

			2	195,40,6		
		R	1	97,50,05		
			2	195,40,4	97,50,4	
A3	A2A3A4	L	1	144,42,36		
			2	289,25,12	144,42,35	144,42,34
		R	1	144,42,34		
			2	289,25,14	144,42,33	
A4	A3A4B1	L	1	116,19,50		
			2	232,39,40	116,19,47	116,19,48
		R	1	116,19,50		
			2	232,39,40	116,19,50	

CO-ORDINATES		
POINT ID	X	Y
B1	-324.9	15.5
A1	-198.5	22.4
B2	-182.5	101.2
A2	-282.6	124.2
A3	-296.4	87.8
A4	-331.4	64.3

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LEARNINGS & TAKE AWAYS:

Ability to adopt high ethical standard in conducting survey project and development tasks.

BOOKS & REFERENCES:

Rangwala,Bhavikatti,Paul.R.Wolf,Donld A Wilson.

DIARY

	ACTIVITY	MEMBERS	ROLL NOS	INSTRUMENTS
DAY-1	SELECTION OF SITE & PERMISSIONS	5		
DAY-2	RECONNAISANCE SURVEY ,COLLECTING TOPO DETAILS	5		
DAY-3				
DAY-4				
DAY-5	RECORDING OF THE DATA IN FMBS & CALCULATIONS			
DAY-6	PLOTTING OF TOPO DETAILS BY AUTOCAD			
DAY-7	TAKING GRID LEVELS AT 20MX20M INTERVALS			
DAY-8				
DAY-9	RECORDING OF THE DATA IN FMBS & CALCULATIONS			
DAY-10	PLOTTING THE DETAILS IN AUTOCAD			
DAY-11	CALCULATIONS FOR DRAIN PROFILING USING EXISTING GRID LEVELS			
DAY-12				
DAY-13	PLOTTING THE DETAILS IN AUTOCAD			
DAY-14	ANGULAR MEASUREMENTS CALCULATION FROM AUTOCAD			
DAY-15	PREPARING PROJECT REPORT			
DAY-16				

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